

Title:**Student Dropout and Academic Success Prediction using Machine Learning****Details:**

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Abstract:

For this project, I used machine learning to study student academic outcomes using the Student Dropout and Success dataset. The dataset included information on student background, admission scores, scholarship status, and semester performance. First I did some exploration of the data set and checked the columns, missing values and overall structure of the data. I also made a couple of graphs to see how different student factors related to final outcomes. Later, Decision Tree and K Nearest Neighbour (KNN) models were trained and evaluated. The comparison of the results of the two models was done with the accuracy and classification metrics to know which model is better for the dataset.

Introduction:

Student dropouts are a common problem for many colleges and universities. Some students drop out of school for financial reasons, academic difficulties or personal reasons. Consequently, the prediction of student performance has become an important topic in data analysis and machine learning. For this project, the Student Dropout and Success dataset was employed to see various factors related to students and how they influence their academic results. The dataset was cleaned and visualised in several ways to analyse it before building the models. In this project, two classification algorithms, Decision Tree and K-Nearest Neighbours (KNN) were used to predict the students' results and compare their performances using evaluation metrics.

Dataset Description:

The dataset used in this project contains information related to student academic performance and background details. It includes features such as admission grades, marital status, scholarship holder status, tuition fee status, gender, age at enrollment, and semester academic records. The target column in the dataset represents the final student outcome, which includes dropout, enrolled, and graduate categories. After loading the dataset into Jupyter Notebook using Pandas, the structure of the dataset was checked using functions such as `df.info()`, `df.shape()`, and `df.describe()`. The dataset was then prepared for further preprocessing and machine learning analysis.

The dataset contains student-related academic and demographic information.

Dataset Shape:

```
[50]: df.shape
[50]: (4424, 37)
```

The dataset contains 4424 rows and 37 columns.

Dataset Information:

```
[16]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4424 entries, 0 to 4423
Data columns (total 37 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Marital status                           4424 non-null   int64
1   Application mode                           4424 non-null   int64
2   Application order                         4424 non-null   int64
3   Course                                    4424 non-null   int64
4   Daytime/evening attendance               4424 non-null   int64
5   Previous qualification                    4424 non-null   int64
6   Previous qualification (grade)           4424 non-null   float64
7   Nacionality                              4424 non-null   int64
8   Mother's qualification                   4424 non-null   int64
9   Father's qualification                   4424 non-null   int64
10  Mother's occupation                      4424 non-null   int64
11  Father's occupation                      4424 non-null   int64
12  Admission grade                          4424 non-null   float64
13  Displaced                               4424 non-null   int64
14  Educational special needs                4424 non-null   int64
15  Debtor                                  4424 non-null   int64
16  Tuition fees up to date                  4424 non-null   int64
17  Gender                                   4424 non-null   int64
18  Scholarship holder                       4424 non-null   int64
19  Age at enrollment                        4424 non-null   int64
20  International                            4424 non-null   int64
21  Curricular units 1st sem (credited)      4424 non-null   int64
22  Curricular units 1st sem (enrolled)      4424 non-null   int64
23  Curricular units 1st sem (evaluations)   4424 non-null   int64
24  Curricular units 1st sem (approved)      4424 non-null   int64
25  Curricular units 1st sem (grade)         4424 non-null   float64
26  Curricular units 1st sem (without evaluations) 4424 non-null   int64
27  Curricular units 2nd sem (credited)      4424 non-null   int64
28  Curricular units 2nd sem (enrolled)      4424 non-null   int64
29  Curricular units 2nd sem (evaluations)   4424 non-null   int64
```

The dataset contains both numerical and categorical features related to student academic performance.

Missing Values:

```
df.isnull().sum()

Marital status                                0
Application mode                              0
Application order                             0
Course                                         0
Daytime/evening attendance                    0
Previous qualification                         0
Previous qualification (grade)                 0
Nacionality                                   0
Mother's qualification                        0
Father's qualification                        0
Mother's occupation                           0
Father's occupation                           0
Admission grade                              0
Displaced                                     0
Educational special needs                     0
Debtor                                         0
Tuition fees up to date                       0
Gender                                         0
Scholarship holder                           0
Age at enrollment                            0
International                                 0
Curricular units 1st sem (credited)           0
Curricular units 1st sem (enrolled)           0
Curricular units 1st sem (evaluations)         0
Curricular units 1st sem (approved)           0
Curricular units 1st sem (grade)              0
Curricular units 1st sem (without evaluations) 0
Curricular units 2nd sem (credited)           0
Curricular units 2nd sem (enrolled)           0
Curricular units 2nd sem (evaluations)         0
Curricular units 2nd sem (approved)           0
Curricular units 2nd sem (grade)              0
Curricular units 2nd sem (without evaluations) 0
Unemployment rate                             0
```

Missing values were checked before preprocessing the dataset.

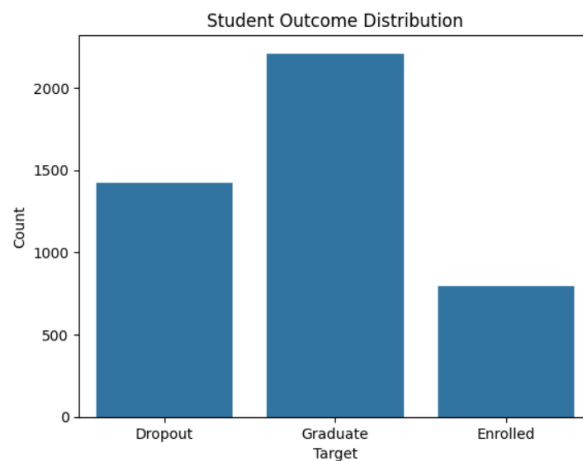
Data Preprocessing

Prior to training the machine learning models , the dataset was preprocessed to make the data suitable for analysis . The first step was to clean up the column names of the dataset from unwanted spaces and special characters. Next, the target column with categorical values such as dropout, enrolled, and graduate was converted to numerical format using label encoding. The dataset was then split into feature variables and target variables. The data was prepared for training and testing by applying the train-test split with 80:20 ratio. Finally, StandardScaler was used to perform feature scaling to normalise the numerical values especially to improve the performance of the KNN model.

Exploratory Data Analysis (EDA)

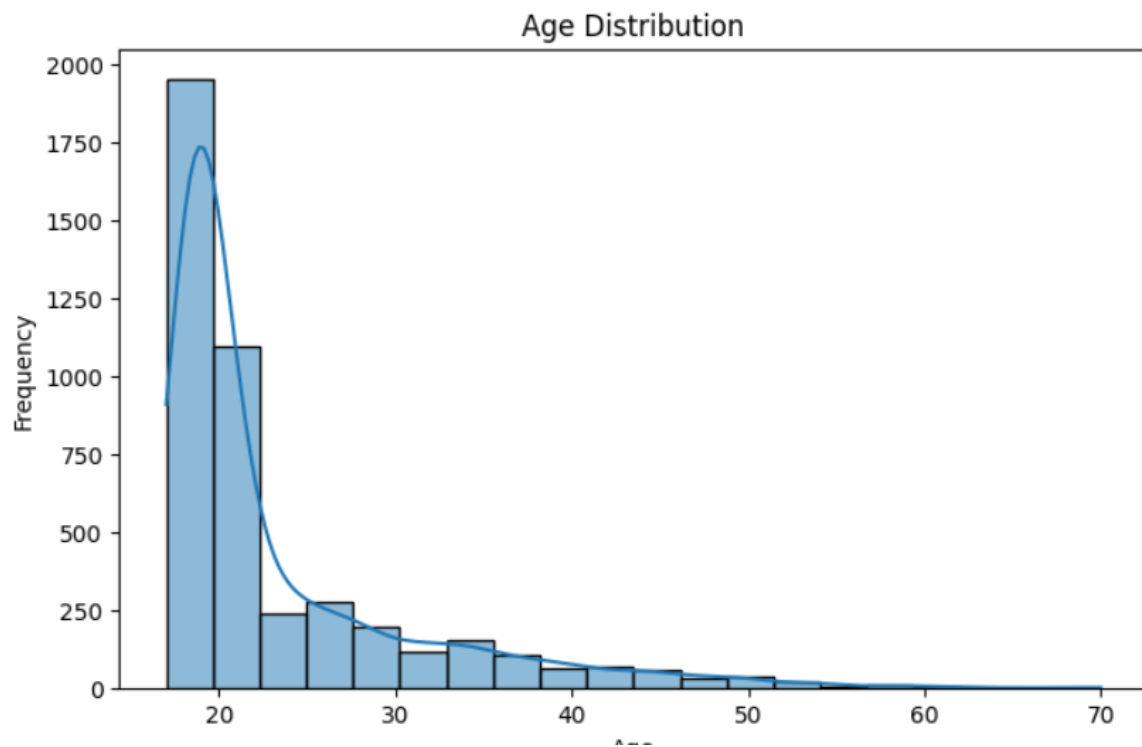
Exploratory Data Analysis was conducted to better understand the dataset and to find patterns related to student outcomes. Various graphs and visualisations were made with Matplotlib and Seaborn libraries. Graph of the target distribution helped to understand the number of students in dropout, enrolled and graduate categories. Other visualisations such as age distribution, gender comparison, admission grade analysis, scholarship status and tuition fee status were also studied to investigate the relationship between the student performance. A correlation heatmap was created to visualise the correlation between the numerical features in the dataset. These visualisations helped in identifying the important factors which may have an impact on student academic success and dropout behaviour.

```
[21]: sns.countplot(x='Target', data=df)
plt.title("Student Outcome Distribution")
plt.xlabel("Target")
plt.ylabel("Count")
plt.show() #getting the student outcome distribution
```



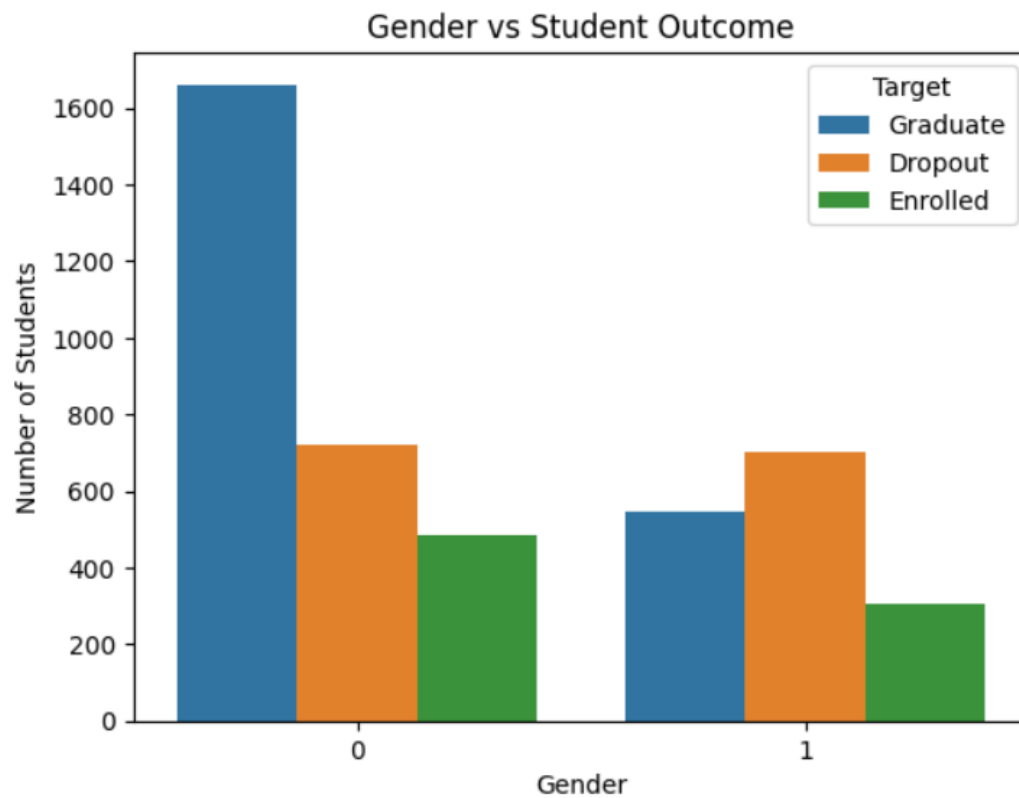
The graph shows the number of students in dropout, enrolled, and graduate categories.

```
] : plt.figure(figsize=(8,5))
sns.histplot(df['Age at enrollment'], bins=20, kde=True)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show() #getting the age dsitribution graph
```



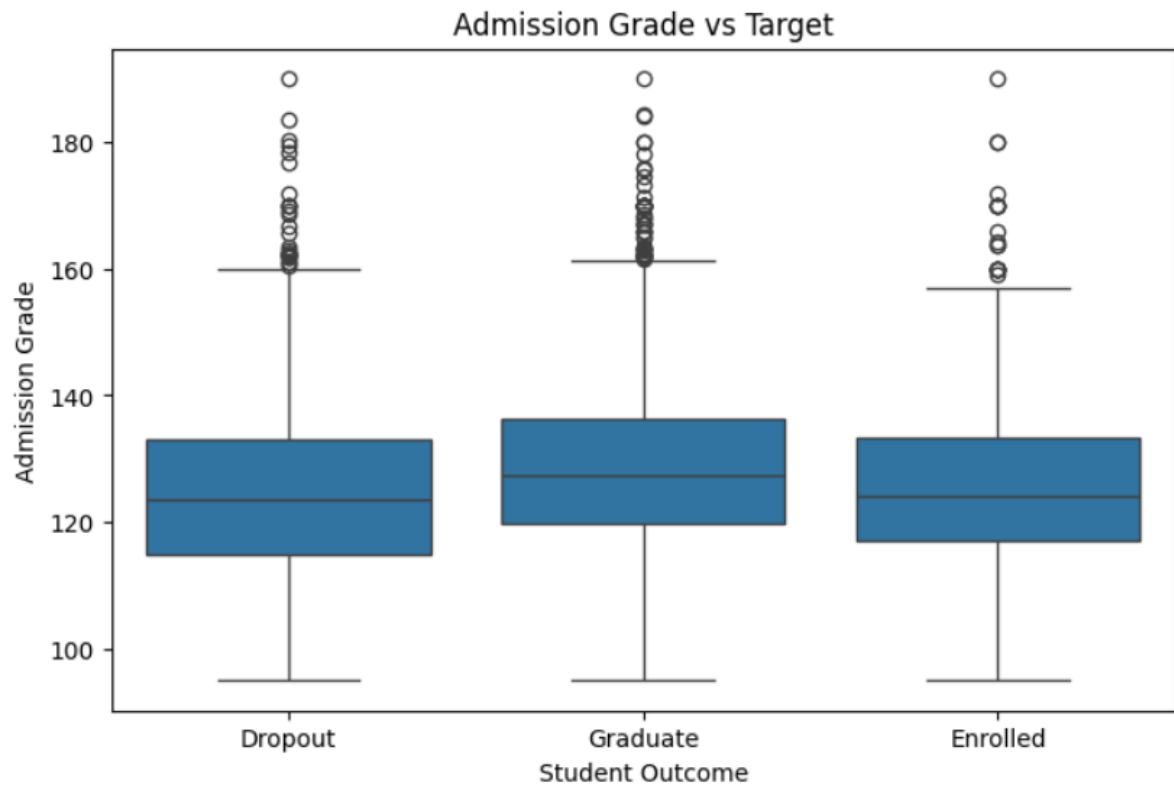
Most students enrolled at younger ages compared to older age groups.

```
#now let us get gender vs student outcome
sns.countplot(x='Gender', hue='Target', data=df)
plt.title("Gender vs Student Outcome")
plt.xlabel("Gender")
plt.ylabel("Number of Students")
plt.show()
```



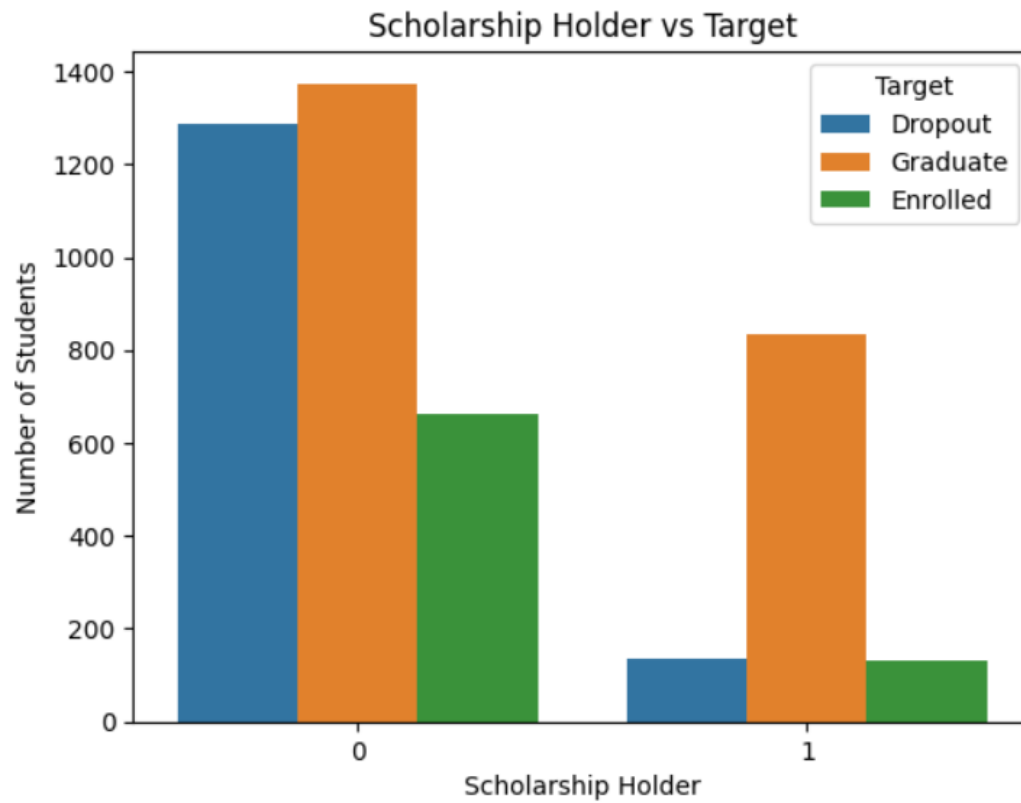
The graph compares student academic outcomes based on gender.

```
#after that we can get te admission grade vs ttarget
plt.figure(figsize=(8,5))
sns.boxplot(x='Target', y='Admission grade', data=df)
plt.title("Admission Grade vs Target")
plt.xlabel("Student Outcome")
plt.ylabel("Admission Grade")
plt.show()
```



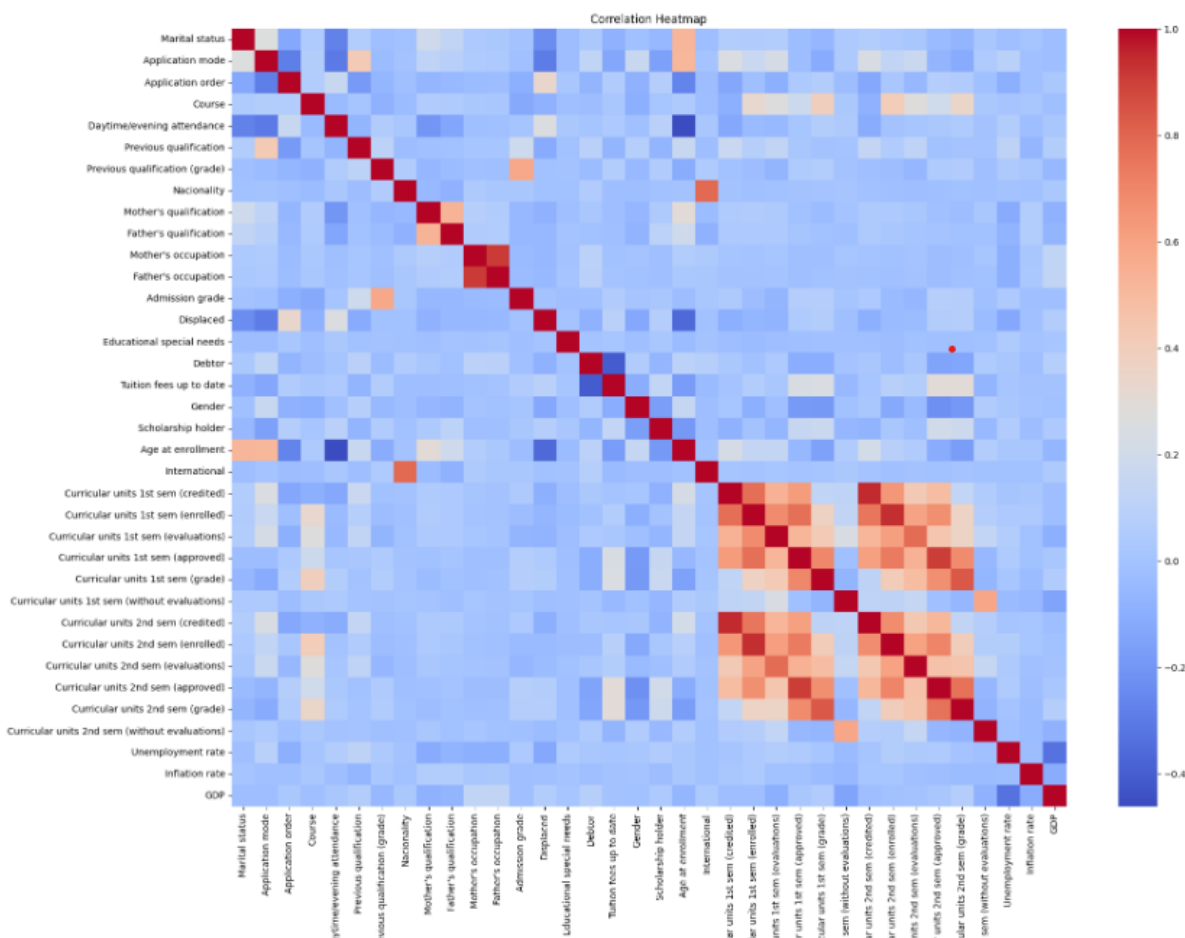
Students with higher admission grades generally showed better academic outcomes.

```
sns.countplot(x='Scholarship holder', hue='Target', data=df)
plt.title("Scholarship Holder vs Target")
plt.xlabel("Scholarship Holder")
plt.ylabel("Number of Students")
plt.show()
```



Scholarship holders showed lower dropout rates compared to non-scholarship students.

```
plt.figure(figsize=(20,15))
sns.heatmap(df.corr(numeric_only=True), cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show() #correaltion heatmap
```



The heatmap shows relationships between numerical variables in the dataset.

Decision Tree Model

For this project, the first machine learning model was the Decision Tree classification algorithm. This model uses decision rules with different feature values to classify the target variable. The dataset was preprocessed and divided into training and testing sets. The Decision Tree model was trained using the training data and evaluated on the testing data. Model performance was evaluated using Accuracy Score, Confusion matrix and Classification report. The model was able to classify student outcome well, and gave good prediction results for the dataset used in this project.

```
from sklearn.metrics import confusion_matrix
print(confusion_matrix(y_test, dt_pred))
```

```
#decision tree evaluation
```

```
[[201  69  46]
 [ 36  61  54]
 [ 41  41 336]]
```

```
from sklearn.metrics import classification_report
print(classification_report(y_test, dt_pred))
```

#classification report

	precision	recall	f1-score	support
0	0.72	0.64	0.68	316
1	0.36	0.40	0.38	151
2	0.77	0.80	0.79	418
accuracy			0.68	885
macro avg	0.62	0.61	0.61	885
weighted avg	0.68	0.68	0.68	885

K-Nearest Neighbours (KNN) Model

The second classification model used in this project was the K-Nearest Neighbours (KNN) algorithm. KNN classifies the data point according to the nearest neighbouring observations. Since KNN is sensitive to the differences in feature values, feature scaling was performed using StandardScaler before applying the KNN model. The training dataset was used to train the model and the testing dataset was used to test the trained model. Its performance was assessed using accuracy score, confusion matrix and classification report. KNN model results were compared with Decision Tree model to know the better algorithm for predicting student academic outcome.

```
from sklearn.neighbors import KNeighborsClassifier
knn_model = KNeighborsClassifier(n_neighbors=5)
knn_model.fit(X_train, y_train)
knn_pred = knn_model.predict(X_test)
knn_accuracy = accuracy_score(y_test, knn_pred)
print("KNN Accuracy:", knn_accuracy)
```

#training knn model

KNN Accuracy: 0.7016949152542373

```
print(confusion_matrix(y_test, knn_pred))
```

#knn evaluation

```
[[216  39  61]
 [ 29  43  79]
 [ 22  34 362]]
```

```
print(classification_report(y_test, knn_pred))
```

```
#classification report
```

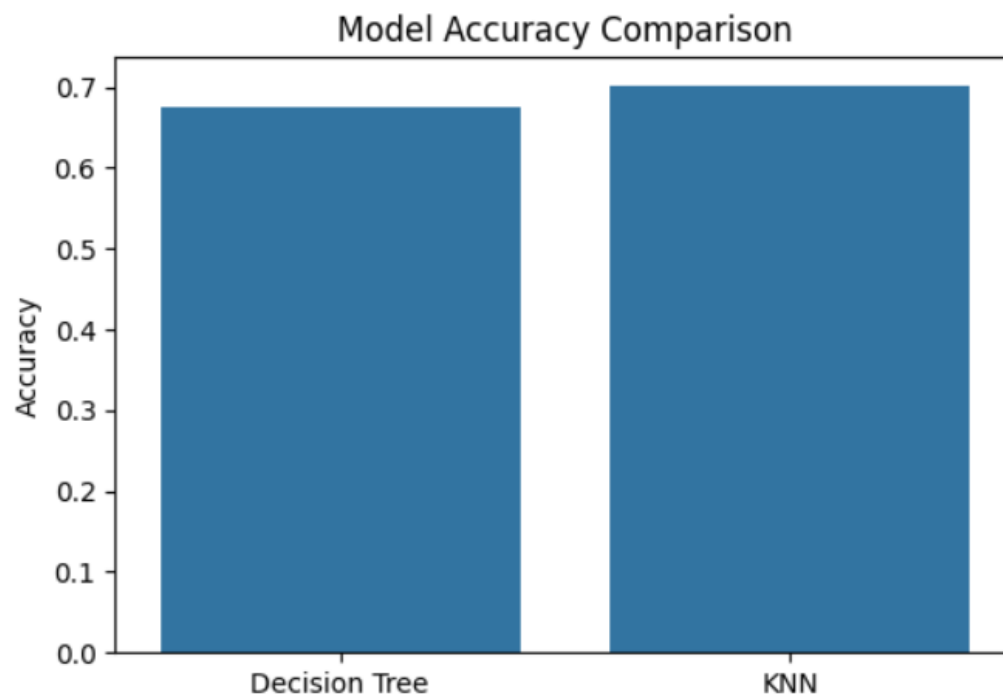
	precision	recall	f1-score	support
0	0.81	0.68	0.74	316
1	0.37	0.28	0.32	151
2	0.72	0.87	0.79	418
accuracy			0.70	885
macro avg	0.63	0.61	0.62	885
weighted avg	0.69	0.70	0.69	885

Model Comparisons

Both the machine learning models were trained and tested and their performances were compared using evaluation metrics such as accuracy score, confusion matrix and classification report . The dataset prediction results of the Decision Tree and K-Nearest Neighbours (KNN) models are good. But the testing results show that one model is a little better than the other in accuracy. In this classification task, the comparison helped to understand the strengths and weaknesses of both algorithms. Finally, the model performance was evaluated to find out which algorithm is better to predict student academic outcome using this dataset.

```
#Accuracy comparision graph

models = ['Decision Tree', 'KNN']
accuracies = [dt_accuracy, knn_accuracy]
plt.figure(figsize=(6,4))
sns.barplot(x=models, y=accuracies)
plt.title("Model Accuracy Comparison")
plt.ylabel("Accuracy")
plt.show()
```



The graph compares the performance of Decision Tree and KNN models.

Feature Importance:

```
#feature importance graph

importance = dt_model.feature_importances_
feature_importance = pd.DataFrame({
    'Feature': X.columns,
    'Importance': importance
})
feature_importance = feature_importance.sort_values(
    by='Importance',
    ascending=False
)
print(feature_importance.head(10))
```

	Feature	Importance
30	Curricular units 2nd sem (approved)	0.356199
12	Admission grade	0.046509
16	Tuition fees up to date	0.043740
6	Previous qualification (grade)	0.042306
31	Curricular units 2nd sem (grade)	0.038275
19	Age at enrollment	0.037593
23	Curricular units 1st sem (evaluations)	0.032376
22	Curricular units 1st sem (enrolled)	0.031806
25	Curricular units 1st sem (grade)	0.030892
29	Curricular units 2nd sem (evaluations)	0.030004

The graph shows the most important features affecting student academic outcomes.

Conclusion:

This project is about student academic outcome prediction using machine learning techniques. Analysis of Student Dropout and Success dataset has been performed using pre-processing techniques and Exploratory Data Analysis techniques to have a better understanding of the data. The relationship between prominent student-related variables and academic performance was examined by creating diverse visual representations. Two classification models Decision Tree and K-Nearest Neighbours (KNN) were trained and validated with different performance metrics. Results indicated that both models were good at predicting students' outcomes. The project also provided insight into how academic performance, financial status and student background influence dropout and graduation behaviour.

References

1. Scikit-learn Documentation. Available at: <https://scikit-learn.org/>
2. Pandas Documentation. Available at: <https://pandas.pydata.org/>
3. Seaborn Documentation. Available at: <https://seaborn.pydata.org/>
4. Matplotlib Documentation. Available at: <https://matplotlib.org/>
5. Student Dropout and Academic Success Dataset. UCI Machine Learning Repository.